

# R Pranav Ajay

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## Education

### Amrita School of Engineering

Bachelor of Technology Computer Science & Engineering *GPA: 8.04*

### Maharishi International Residential School

Higher Secondary, CBSE Computer Science *GPA: XII- 455/500, X- 10 CGPA*

Ettimadai, Coimbatore

Jul 2018 – Jun 2022

Sriperumbudur, Kanchipuram

2016 – 2018

## Experience

### Caterpillar

Bengaluru, Karnataka, India

*Senior Associate Engineer*

Aug 2024 – Present

- Spearheaded optimization, deterministic profiling, and memory management of core C++ application frameworks, aggressively reducing binary sizes for deployment onto resource-constrained automotive Electronic Control Modules (ECMs).
- Developed and integrated communication and physics-accurate simulation pipelines into Linux-based Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) infrastructure utilizing custom Project Chrono and Unreal Engine forks.
- Implemented system co-simulation synchronization layers, executing low-level telemetry validation and hardware integration testing via CAN, NMEA, UART, MODBUS RTU, and DPX (Ranging Nodes).
- Engineered end-to-end automated testing configurations and CI/CD pipelines using Bash/Shell scripts to optimize multi-threaded deployment speed.
- Represented the regional engineering team on an international technical deployment to Brazil to audit core performance metrics and align system requirements.

*Associate Engineer*

Aug 2022 – Aug 2024

- Implemented optimized scene graph architectures, object state handlers, and custom asset evaluation pipelines for concurrent simulation frameworks.
- Designed real-time event-driven scenario generators utilizing discrete event simulation concepts (SimPy frameworks) wrapped in high-performance C++ execution logic.
- Utilized advanced data structures and algorithms to handle spatial state transformations and high-frequency communication packet routing under Linux environments.

*College Intern*

Jan 2022 – Jul 2022

- Maintained and updated application layer frameworks across proprietary simulation environments using modular scene graph architectures.
- Gained comprehensive engineering exposure to native C++ build configurations and system-level toolchains including Qt, CMake, Clang, GCC, and vcpkg.
- Automated raw multi-body asset ingestion and scene-graph generation passes to streamline testing routines.

## Skills

**Programming Languages:** C++ (Up to C++17/20), Python, Java, Kotlin, Bash/Shell Scripting, JavaScript

**Tools & Protocols:** Linux (Fedora/Ubuntu), Git, CMake, CAN, NMEA, UART, DPX, MODBUS RTU, ADB, Fastboot

**Frameworks & Systems:** AOSP Environment, Android Studio, Boost, Poco, Dear ImGui, Qt, TensorFlow Lite

## Projects

### AOSP Custom ROM Integration (Palladium OS)

*Kotlin, Android Studio, Framework Modding*

[palladiumos archive](#)

- Developed 'Atomic Hub', a centralized system customization Android application natively integrated into the AOSP Settings framework.
- Architected the application frontend and UI/UX behaviors, directly interfacing user inputs with the backend core framework logic to execute system-level OS configurations.

### ToonLytV2 (Edge Device ML Applications)

*TensorFlow Lite, Kotlin, Android SDK*

[github.com/sairpa/ToonLytV2](https://github.com/sairpa/ToonLytV2)

- Developed a custom Android application that optimizes real-time edge workloads by running local style-transfer networks.
- Managed application memory constraints, background thread lifecycle behaviors, and asset pipelines within native Android runtime restrictions.

### Kreative Rum & Technical Pipeline Infrastructure

*C++, CMake, Build Automation*

[github.com/sairpa/kreative\\_rum](https://github.com/sairpa/kreative_rum)

- Built a centralized sandbox demonstrating custom tools, modular library linking, and build automation pipelines optimized for real-time asset compilation and resource tracking.